

VPN Setup Guide

SSL/TLS → SSH Tunnel with Custom Payload
Ubuntu VPS — Step-by-Step Documentation

Architecture: Stunnel (SSL/TLS) → OpenSSH → Custom HTTP Payload → HTTP Injector (.ehi)

Ubuntu 20.04 / 22.04 LTS

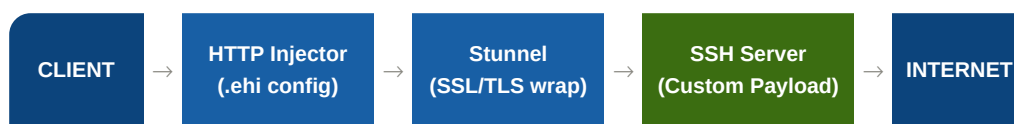
Stunnel4

OpenSSH

HTTP Injector

Version 1.0

ARCHITECTURE OVERVIEW



1

VPS Preparation

Update system and install required packages

1.1 — Connect to your VPS via SSH

From your local machine, open a terminal and connect:

```
ssh root@YOUR_VPS_IP
# Replace YOUR_VPS_IP with your actual server IP address
```

1.2 — Update & upgrade the system

```
apt update && apt upgrade -y
apt install -y curl wget nano ufw
```

1.3 — Set the server hostname (optional but recommended)

```
hostnamectl set-hostname vpn-server
```

NOTE: Always run apt update before installing packages to ensure you get the latest versions.

2

OpenSSH Server Setup

Configure SSH for tunneling with custom payload support

2.1 — Install OpenSSH Server

```
apt install -y openssh-server
systemctl enable ssh
systemctl start ssh
systemctl status ssh # Should show: active (running)
```

2.2 — Configure SSH settings

Edit the SSH daemon configuration file:

```
nano /etc/ssh/sshd_config
```

Find and set (or add) the following lines:

```
Port 22
PermitRootLogin yes
PasswordAuthentication yes
GatewayPorts yes
AllowTcpForwarding yes
X11Forwarding no
ClientAliveInterval 60
ClientAliveCountMax 3
```

NOTE: For production servers, consider disabling PasswordAuthentication and using SSH keys instead.

2.3 — Create a dedicated VPN user

```
adduser vpnuser
# Enter a strong password when prompted

# Optionally restrict the user to SSH tunneling only:
usermod -s /bin/false vpnuser
```

2.4 — Restart SSH to apply changes

```
systemctl restart ssh
```

3

Stunnel SSL/TLS Layer

Wrap SSH traffic inside SSL/TLS encryption

3.1 — Install Stunnel

```
apt install -y stunnel4
```

3.2 — Generate a self-signed SSL certificate

```
openssl req -new -newkey rsa:2048 -days 3650 -nodes -x509 \
-subj '/CN=vpn-server' \
-keyout /etc/stunnel/stunnel.key \
-out /etc/stunnel/stunnel.crt

# Combine into a single PEM file
cat /etc/stunnel/stunnel.key /etc/stunnel/stunnel.crt \
> /etc/stunnel/stunnel.pem
chmod 600 /etc/stunnel/stunnel.pem
```

3.3 — Create Stunnel configuration

```
nano /etc/stunnel/stunnel.conf
```

Paste the following configuration:

```
; /etc/stunnel/stunnel.conf
cert = /etc/stunnel/stunnel.pem
socket = l:TCP_NODELAY=1
socket = r:TCP_NODELAY=1
output = /var/log/stunnel4/stunnel.log

[ssh-ssl]
accept = 443 ; SSL/TLS port clients connect to
connect = 127.0.0.1:22 ; Forward to local SSH
```

3.4 — Enable and start Stunnel

```
# Enable in /etc/default/stunnel4
sed -i 's/ENABLED=0/ENABLED=1/' /etc/default/stunnel4

systemctl enable stunnel4
systemctl start stunnel4
systemctl status stunnel4 # Should show: active (running)
```

4

Firewall Configuration

Open required ports using UFW

4.1 — Configure UFW firewall rules

```
ufw allow 22/tcp # SSH (direct)
ufw allow 443/tcp # SSL/TLS (Stunnel)
ufw allow 80/tcp # HTTP (for payload injection)
ufw allow 8080/tcp # Alternative HTTP port

ufw enable
ufw status verbose
```

IMPORTANT: If you are connected via SSH, make sure port 22 is allowed BEFORE enabling UFW to avoid locking yourself out.

5

HTTP Injector (.ehi) Configuration

Set up the client-side HTTP Injector app

5.1 — HTTP Injector settings

Open the HTTP Injector app on your Android device and configure:

Setting	Value
Server Host	YOUR_VPS_IP
Server Port	443 (Stunnel SSL port)
SSH Host	127.0.0.1 (via Stunnel tunnel)
SSH Port	22
SSH Username	vpnuser
SSH Password	[your vpnuser password]
Connection Mode	SSL/TLS
SNI / Payload Host	YOUR_VPS_IP or domain

5.2 — Custom HTTP Payload

In the Payload section of HTTP Injector, use this template:

```
CONNECT [host_port] HTTP/1.1[crLf]
Host: YOUR_VPS_IP[crLf]
Upgrade: websocket[crLf]
Connection: Upgrade[crLf][crLf]
```

NOTE: Replace YOUR_VPS_IP with your actual server IP. [crLf] is automatically inserted by HTTP Injector — type it literally.

6

Testing & Verification

Confirm all services are running correctly

6.1 — Check all services on the VPS

```
# Check SSH
systemctl status ssh

# Check Stunnel
systemctl status stunnel4

# Verify listening ports
ss -tlnp | grep -E '22|443|80'

# Test SSL connection locally
openssl s_client -connect 127.0.0.1:443
```

6.2 — Test SSH through Stunnel from a remote machine

```
# Install stunnel on your local Linux/Mac machine
apt install stunnel4 # or brew install stunnel on Mac

# Connect through SSL tunnel to test
ssh -o ProxyCommand='openssl s_client -quiet -connect YOUR_VPS_IP:443' \
vpnuser@YOUR_VPS_IP
```

6.3 — Connection checklist

- ✓ SSH service is running on port 22
- ✓ Stunnel is running and listening on port 443
- ✓ UFW firewall allows ports 22, 80, 443
- ✓ HTTP Injector connects and shows 'Connected' status
- ✓ Internet traffic routes through the VPN (verify IP change)
- ✓ Connection remains stable for 5+ minutes

7

Backup & Maintenance

Keep your configuration safe and the server running smoothly

7.1 — Back up your configuration files

```
# Create a backup archive of all config files
tar -czvf vpn_config_backup_$(date +%Y%m%d).tar.gz \
/etc/ssh/sshd_config \
/etc/stunnel/ \
/etc/default/stunnel4

# Download to your local machine
scp root@YOUR_VPS_IP:~/vpn_config_backup_*.tar.gz ./
```

7.2 — Useful maintenance commands

```
# Restart all VPN services
systemctl restart ssh stunnel4

# View Stunnel connection log
tail -f /var/log/stunnel4/stunnel.log

# View active SSH sessions
who
ss -tnp | grep ssh

# Check server uptime and load
uptime
```

QUICK REFERENCE

Component	Port	Config File	Service Command
SSH Server	22	/etc/ssh/sshd_config	systemctl restart ssh
Stunnel (SSL)	443	/etc/stunnel/stunnel.conf	systemctl restart stunnel4
SSL Certificate	—	/etc/stunnel/stunnel.pem	openssl req ... (regenerate)
Firewall (UFW)	—	ufw rules	ufw status verbose
VPN User	—	/etc/passwd	adduser vpnuser

TROUBLESHOOTING

Cannot connect on port 443	Check UFW rules (ufw allow 443/tcp) and confirm Stunnel is running
SSH connection refused	Verify SSH service is active: systemctl status ssh
Stunnel fails to start	Check certificate path in stunnel.conf and run: stunnel /etc/stunnel/stunnel.conf
HTTP Injector shows 'Timeout'	Verify Server Host/Port match VPS IP and Stunnel port (443)
Connection drops frequently	Increase ClientAliveInterval in sshd_config and check server load
Certificate errors on client	Regenerate the PEM file or set 'verify = 0' in client stunnel config